Module 2: Dataset Notes

No-Code AI Lab: Build, Train, and Test Without Coding

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Purpose: These notes summarize my initial exploration of a small, labeled image dataset and the preparation steps needed before evaluating an image-classification model.

Class	Image samples
Pen	3
Pencil	3
Total	6

1. Dataset Overview

- **Dataset type:** Image dataset for a no-code image-classification experiment.
- **What it contains:** Two classes: Pen and Pencil, with three image samples in each class (six samples total).
- **Goal:** Explore how an AI model can learn from labeled images and classify a new image as a pen or a pencil.

2. Structure and Features

- **Organization:** The images are grouped into two labeled classes in Google Teachable Machine: Pen and Pencil.
- **Inputs:** Images showing pens and pencils. The model uses visual information in the images as its input.
- **Labels:** Pen and Pencil. Each label identifies the object category that the model is expected to learn.
- **Observations:** The classes have equal numbers of samples, but the dataset is very small. If images share similar backgrounds, angles, or lighting, the model may not learn features that generalize well to new images.

3. Preparation Steps

- **Completed during initial exploration:** Created/identified the Pen and Pencil classes and organized the available image examples under the relevant labels in Teachable Machine.
- **Checks to carry out:** Review each image to confirm its label is correct; remove duplicate, blurry, irrelevant, or incorrectly labeled images; and use clear, consistently labeled examples.
- **Next preparation steps:** Collect more images with varied angles, backgrounds, object appearances, and lighting. Keep separate unseen images for testing model performance. The current dataset contains only three examples per class, so more samples are needed for a meaningful evaluation.

4. Bias Awareness

- **Balance:** The dataset is numerically balanced: three Pen images and three Pencil images.
- **Possible gaps:** Equal class counts do not guarantee variety. The dataset may not represent different pen and pencil designs, colors, sizes, backgrounds, camera angles, or lighting conditions.
- **Potential impact:** The model could learn incidental background or lighting patterns rather than reliable differences between pens and pencils. It may then misclassify unfamiliar images.
- **How to reduce the risk:** Add diverse examples to both classes, check labels carefully, and test with images that were not used for training.

Reflection

This activity helped me understand that dataset quality is not only about the number of examples. Clear labels, variety, and representative samples are also important. Before drawing conclusions about model performance, I need to expand the

dataset and test the model using new images.

Note: This document records the initial dataset review and planned quality checks. It does not claim that additional images have already been collected or that model accuracy has been measured.